

GTK AEROGEOPHYSICAL SURVEY DESCRIPTION

The survey includes three survey methods:

- Magnetic total field measurements
- Electromagnetic measurements
- Radiometric gamma-ray measurements

The detailed description of survey equipment systems is proposed in appendix 1. The files were compressed to zip –files (ZipGenius6) so that each zip –file represent one flight area. The index map is presented in appendix 2. The content of appendix 2 is represented as ArcGis shape –file where the data attributes are as follow:

AREA	Name of flight area
YEAR	Flight year
AIRCRAFT	Aircraft (KOG = Twin Otter, VKB = DC-3, USI = Cessna Caravan)
DIRECTION	Flight direction
SPACING	Flight line spacing
NUMBER	Number of magnetometers
TYPE	Type of magnetometer (C = cesium, P = proton)
FREQUENCY	Number of AEM frequencies

GTK DATA FORMAT DESCRIPTION

XYZ Files

There is one Geosoft XYZ file corresponding to each survey method (magnetic, radiometric and electromagnetic measurements). Coordinate system based on Gauss-Krueger projection and central meridian of 21°E for zone 2 map sheets (KKJ2), 27°E for zone 3 map sheets (KKJ3) and 30°E for zone 4 map sheets (KKJ4).

Electromagnetic coil configuration was changed in 1980 from vertical coaxial to vertical coplanar. In vertical coaxial coil configuration system conductors are located as negative values and on the contrary as positive values in vertical coplanar system.

Apparent resistivity and depth was calculated from primary EM components by a half-space model.

The files from the year of 1973 to 1995 are named as follows (# means the first three letters of the area name):

AP#.XYZ	apparent resistivity data
EM#.XYZ	electromagnetic data
MB#.XYZ	back boom magnetometer data
ML#.XYZ	left wingtip magnetometer data
MR#.XYZ	right wingtip magnetometer data
RA#.XYZ	radiometric data

The files from the year of 1996 to 2004 are named respectively:

AP#.XYZ	apparent resistivity data
EM#.XYZ	electromagnetic data
ML#.XYZ	left wingtip magnetometer data
MR#.XYZ	right wingtip magnetometer data
RA#.XYZ	radiometric data

The files from the year of 2006 to 2007 are named respectively:

AP#.XYZ	apparent resistivity data
EM#.XYZ	electromagnetic data
ML#.XYZ	left wingtip magnetometer data
MN#.XYZ	nose boom magnetometer data
RA#.XYZ	radiometric data

XYZ File Columns

Each row in these ASCII files corresponds to one measurement point. One column of the row corresponds to one measurement parameter.

The data includes the following columns:

Electromagnetic data files from the year of 1973 to 1995.

X, Y	Easting, northing (meters)
RE	In-phase component (ppm)
IM	Quadrature component (ppm)
RALT	Radar altitude (meters)

Apparent resistivity data from the year of 1973 to 1995

X, Y	Easting, northing (meters)
LFA	Apparent resistivity (Ohm-m)
LFS	Apparent depth to conductor (m)

Electromagnetic data files from the year of 1996 to 2004

X, Y	Easting, northing (meters)
DAY	Day number from the beginning of the year
TIME	Measurement time stamp (hhmmss)
DIR	Flight direction (degrees, clockwise from north)
RALT	Radar altitude (meters)
LFR	In-phase component of the lower frequency (3 125 Hz) (ppm)
LFI	Quadrature component of the lower frequency (3 125 Hz) (ppm)
HFR	In-phase component of the higher frequency (14 368 Hz) (ppm)
HFI	Quadrature component of the higher frequency (14 368 Hz) (ppm)

Apparent resistivity data from the year of 1996 to 2004

X, Y	Easting, northing (meters)
LFA	Apparent resistivity, 3 kHz (Ohm-m)
LFS	Apparent depth to conductor, 3 kHz (m)
HFA	Apparent resistivity, 14 kHz (Ohm-m)
HFS	Apparent depth to conductor, 14 kHz (m)

Electromagnetic data files from the year of 2004 to 2007

X, Y	Easting, northing (meters)
DAY	Day number from the beginning of the year
FLIGHT	Flight material number
TIME	Measurement time stamp (hhmmss)
DIR	Flight direction (degrees, clockwise from north)
RALT	Radar altitude (meters)
GPS_H	GPS altitude above WGS84 ellipsoid

DTM	Digital terrain model (above WGS-84 ellipsoid)
SPHER	Spherics monitor
PLM	Power line monitor (on linear range 0 – 999)
RE09	In-phase component, 0.9 kHz (ppm)
IM09	Quadrature component, 0.9 kHz (ppm)
RE3	In-phase component, 3 kHz (ppm)
IM3	Quadrature component, 3 kHz (ppm)
RE12	In-phase component, 12 kHz (ppm)
IM12	Quadrature component, 12 kHz (ppm)
RE25	In-phase component, 25 kHz (ppm)
IM25	Quadrature component, 25 kHz (ppm)

Apparent resistivity data from the year of 2006 to 2007

X, Y	Easting, northing (meters)
AR09	Apparent resistivity, 0.9 kHz (Ohm-m)
AD09	Apparent depth to conductor, 0.9 kHz (m)
AR3	Apparent resistivity, 3 kHz (Ohm-m)
AD3	Apparent depth to conductor, 3 kHz (m)
AR12	Apparent resistivity, 12 kHz (Ohm-m)
AD12	Apparent depth to conductor, 12 kHz (m)
AR25	Apparent resistivity, 25 kHz (Ohm-m)
AD25	Apparent depth to conductor, 25 kHz (m)

Magnetic data from the year of 1973 to 1995 are in the following order

Easting (meters)
 Northing (meters)
 Total magnetic field at IGRF65 level (nT)
 Radar altitude (meters)
 Barometric altitude (meters)

Magnetic data files from the year of 1996

X, Y	Easting, northing (meters)
DAY	Day number from the beginning of the year
TIME	Measurement time stamp (hhmmss)
DIR	Flight direction (degrees, clockwise from north)
RALT	Radar altitude (meters)
MGCL	Total magnetic field of the left wingtip magnetometer at IGRF65 level (nT)
MGCR	Total magnetic field of the right wingtip magnetometer at IGRF65 level (nT)
MGCN	Total magnetic field of the nose boom magnetometer at IGRF65 level (nT)

Radiometric data files from the year of 1973 to 1995

X, Y	Easting, northing (meters)
TOT	Total radiation (ur –unit)
KAL	Potassium concentration (% K)
URA	Uranium concentration (ppm equivalent uranium eU)
THO	Thorium concentration (ppm equivalent thorium eTh)
RALT	Radar altitude (meters)

Radiometric data files from the year of 1996 to 2004

X, Y	Easting, northing (meters)
DAY	Day number from the beginning of the year

TIME	Measurement time stamp (hhmmss)
DIR	Flight direction (degrees, clockwise from north)
RALT	Radar altitude (meters)
BALT	Barometric altitude (meters)
TOUT	Temperature outside the aircraft (°C)
TOT	Total radiation (ur –unit)
KAL	Potassium concentration (% K)
URA	Uranium concentration (ppm equivalent uranium eU)
THO	Thorium concentration (ppm equivalent thorium eTh)

Radiometric data files from the year of 2006 to 2007

X, Y	Easting, northing (meters)
FLIGHT	Flight material number
DAY	Day number from the beginning of the year
TIME	Measurement time stamp (hhmmss)
DIR	Flight direction (degrees, clockwise from north)
RALT	Radar altitude (meters)
BALT	Barometric altitude (meters)
TOUT	Temperature outside the aircraft (°C)
GPS_H	
DTM	Digital terrain model (
TOT	Total radiation (ur –unit)
KAL	Potassium concentration (% K)
URA	Uranium concentration (ppm equivalent uranium eU)
THO	Thorium concentration (ppm equivalent thorium eTh)

APPENDIX 1

Aircraft:	Magnetometers			Electromagnetics				Radiometrics	
OH-KOG	Magnetometrit			Sähkömagneettinen				Radiometrinen	
Year	Sensors / Anturit		Registration	Coil distance	Frequency	Moment	Registration	Crystal volume	Channels
Vuosi	P=Proton/C=Cesium		Tallennusväli	Kelaväli	Taajuus	Momentti	Tallennusväli	Kidetilavuus	Kanavia
	N:o	C/P	(1/s)	(m)	(Hz)	(Am*2)	(1/s)	(l)	N:o
1973	1	P	2	26,5	3220	127	2	27,3	36
1974	1	P	2	26,5	3220	127	2	27,3	36
1975	2	P	2	25,8	3220	127	2	27,3	36
1976	2	P	2	25,8	3220	127	2	27,3	36
1977	2	P	2	25,8	3220	127	2	27,3	36
1978	2	P	2	25,0	3220	127	2	27,3	54
1979	2	P	2	25,0	3220	127	2	27,3	54
1980	2	P	2	21,44	3222	105	4	25,0	120
1981	2	P	2	21,36	3113	105	4	25,0	120
1982	2	P	2	21,36	3113	105	4	25,0	120
1983	2	P	2	21,36	3113	105	4	25,0	120
1984	3	P	4	21,36	3113	105	4	25,0	120
1985	3	P	4	21,36	3113	105	4	25,0	120
1986	3	P	4	21,36	3113	105	4	25,0	120
1987	3	P	4	21,36	3113	105	4	25,0	120
1988	3	P	4	21,36	3113	105	4	25,0	120
1989	2	P	4	21,36	3113	105	4	25,0	120
1990	2	P	4	21,36	3113	105	4	25,0	120
1991	2	P	4	21,36	3113	105	4	25,0	120
1992	2	C	4	21,36	3113	105	4	25,0	120
1993	2	C	4	21,36	3113	105	4	25,0	120
1994	2	C	4	21,36	3113	105	4	25,0	120
1995	2	C	4	21,36	3113	105	4	25,0	120
1996	2	C	4	21,36	3125/14368	115/55	4/4	25,0	120
1997	2	C	8	21,36	3125/14368	115/55	4/4	41 (33+8)	256
1998	2	C	8	21,36	3125/14368	115/55	4/4	41 (33+8)	256
1999	2	C	10/8	21,36	3125/14368	115/55	4/4	41 (33+8)	256
2000	2	C	10/8	21,36	3125/14368	115/55	4/4	41 (33+8)	256
2001	2	C	10	21,36	3125/14368	115/55	4/4	41 (33+8)	256
2002	2	C	10	21,36	3125/14368	115/55	4/4	41 (33+8)	256
2003	2	C	10	21,36	3125/14368	115/55	4/4	41 (33+8)	256
2004	2	C	10	21,36	3125/14368	115/55	4/4	41 (33+8)	256
2005	2	C	10	21,36	3125/14368	115/55	4/4	41 (33+8)	256
2005	2	C	10	21,36	3125/14368	115/55	4/4	41 (33+8)	256
2006	2	C	10	21,36	912/3005/11962/24510	115/55	4/4	41 (33+8)	256
2007	2	C	10	21,36	912/3005/11962/24510	115/55	4/4	41 (33+8)	256

	Paikannus	Positioning
1973-1975	kiintopisteet	camera fixpoints
1976-1992	kiintopisteet+doppler	camera fixpoints+doppler
1993-	differentiaali GPS	differential GPS



